

VAASA KLEMETTILA, Finland

WMO: 029570

Lat: 63.0989N

Lon: 21.6394E

Elev: 30

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.5	0.1	-10.2	3.2	-5.5	-4.1	4.4	0.5	26.3	34.3	24.0	33.1	5.0	120	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	77.7	64.2	74.2	62.7	71.0	61.2	66.8	73.6	64.7	71.6	62.7	69.2	9.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	90.2	70.6	62.0	83.1	67.8	59.9	77.2	65.7	31.4	73.4	29.7	71.9	28.3	69.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.6	17.4	-7.2	81.4	11.0	3.6	-15.1	84.0	-21.5	86.1	-27.7	88.2	-35.7	90.8		
			-7.7	68.3	10.5	3.0	-15.3	70.5	-21.4	72.2	-27.3	73.9	-34.9	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	22.2	22.7	27.5	37.6	49.1	56.4	62.7	59.7	52.3	41.6	33.8	27.5		
	DBStd	15.94	10.72	11.25	8.25	6.11	7.47	5.37	5.41	4.72	5.36	6.05	7.42	9.17		
	HDD50	4300	862	765	699	374	107	7	0	1	38	265	486	696		
	HDD65	8750	1327	1185	1164	821	497	262	112	178	382	725	936	1161		
	CDD50	1086	0	0	0	3	77	200	394	301	106	5	0	0		
	CDD65	62	0	0	0	0	3	5	41	13	0	0	0	0		
	CDH74	336	0	0	0	0	24	24	234	53	1	0	0	0		
	CDH80	51	0	0	0	0	1	0	44	6	0	0	0	0		
	WSAvg	8.6	8.8	9.3	9.0	8.4	8.8	8.9	8.1	8.3	7.9	8.5	8.6	9.2		
	PrecAvg	19.9	1.3	0.9	1.0	0.9	1.3	1.7	2.4	2.6	2.3	2.1	1.9	1.6		
Precipitation	PrecMax	27.9	2.8	2.4	2.2	2.6	3.0	4.7	6.0	7.1	4.3	4.0	4.5	3.8		
	PrecMin	13.2	0.2	0.1	0.2	0.0	0.1	0.1	0.2	0.2	0.6	0.7	0.4	0.3		
	PrecStd	3.4	0.7	0.6	0.5	0.6	0.7	0.9	1.6	1.5	1.1	0.9	0.9	0.9		
	PrecStd	3.4	0.7	0.6	0.5	0.6	0.7	0.9	1.6	1.5	1.1	0.9	0.9	0.9		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	42.2	46.4	62.7	77.1	77.1	84.9	79.9	68.9	57.4	50.3	44.1		
		MCWB	38.2	38.1	40.1	48.7	59.8	62.0	66.5	67.7	60.4	52.5	47.1	42.4		
	2%	DB	38.3	39.7	42.5	54.0	71.5	72.9	79.9	74.5	64.9	54.2	46.7	42.1		
		MCWB	36.3	36.9	37.7	45.5	57.0	61.0	65.5	63.9	58.1	51.6	44.7	40.6		
	5%	DB	36.4	37.3	40.1	50.4	66.6	69.2	76.0	71.0	62.4	51.9	44.6	40.2		
		MCWB	34.9	35.4	36.3	43.4	54.7	58.5	64.2	61.5	56.5	49.7	43.1	38.8		
	10%	DB	34.7	35.6	38.0	47.3	61.1	66.1	72.4	67.8	60.2	50.1	43.0	38.1		
		MCWB	33.6	34.2	35.1	41.3	51.6	57.2	63.0	60.4	55.6	47.8	41.8	36.9		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	39.0	40.8	50.9	62.2	65.3	70.0	69.7	61.9	54.4	47.8	43.3		
		MCDB	40.1	41.7	45.7	58.8	73.5	73.2	78.8	77.4	66.5	56.1	50.6	44.0		
	2%	WB	36.7	36.9	38.5	46.4	58.8	62.4	67.7	65.6	59.8	52.0	45.0	41.0		
		MCDB	37.9	38.8	41.4	53.2	68.7	69.0	74.8	71.3	63.3	53.7	46.2	41.9		
	5%	WB	35.1	35.8	37.0	43.9	55.3	60.3	66.0	63.2	57.9	50.2	43.3	38.7		
		MCDB	36.2	37.3	39.5	49.4	65.2	67.0	72.9	69.0	60.9	51.7	44.3	39.8		
	10%	WB	33.8	34.3	35.3	41.7	52.2	58.2	64.1	61.3	56.4	48.2	41.9	36.9		
		MCDB	34.7	35.4	37.7	46.4	59.8	64.7	71.0	66.0	59.4	49.7	43.0	37.9		
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.6	11.7	12.6	13.7	12.5	12.6	12.0	11.0	8.2	6.0	7.5		
		MCDBR	7.0	8.8	10.7	18.8	21.9	18.3	18.4	17.5	14.3	9.2	6.6	7.0		
		MCWBR	6.3	7.6	7.5	11.2	10.5	9.1	7.8	8.4	8.6	7.3	6.3	6.4		
	5% WB	MCDBR	6.8	8.5	9.6	16.4	20.1	15.6	15.6	14.3	11.8	8.2	6.3	6.8		
		MCWBR	6.4	7.5	7.3	10.4	10.1	8.7	7.8	8.0	8.0	7.3	6.2	6.4		
Clear-Sky Solar Irradiance	taub		0.240	0.296	0.319	0.357	0.355	0.353	0.374	0.361	0.335	0.318	0.381	0.350		
	taud		2.064	2.190	2.276	2.319	2.433	2.469	2.433	2.462	2.516	2.381	2.663	2.554		
	Ebn at Noon		122	200	246	261	273	276	266	259	244	195	112	67		
	Edh at Noon		10	20	27	32	31	31	31	28	22	17	9	6		
All-Sky Solar Radiation	RadAvg		46	226	705	1211	1669	1869	1709	1305	766	292	71	23		
	RadStd		6	45	82	122	158	139	150	115	73	56	13	2		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page